

# Deep Learning Models of Loan Portfolios and Asset-Backed Securities



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# Chapter 1

## Data

### 1.1 Data Source

This study employs the Fannie Mae Single-Family Loan Performance (SFLP) dataset, a publicly available research dataset maintained by the Federal National Mortgage Association (Fannie Mae) and updated on a quarterly basis. The dataset was originally released in April 2013 and contains origination and monthly performance records for a subset of single-family conventional mortgage loans acquired by Fannie Mae since January 2000. New loan acquisitions and performance data are added with an approximate four-month lag following the end of each calendar quarter fanniema2024.

The data are organised as a *loan-month panel*: each observation corresponds to a single mortgage loan in a single monthly reporting period, beginning at the month of acquisition and continuing until the loan terminates through full prepayment, foreclosure, Real Estate Owned (REO) disposition, or contractual maturity, or until the data cut-off date. This longitudinal structure is well-suited to the competing-risks framing of sirignano2021, which the present study adapts to the Fannie Mae data.

Raw files are distributed as pipe-delimited CSV without a header row, organised by acquisition quarter and identified by the convention `YYYYQn.csv`. Column identities are positional, fully specified in the Fannie Mae Single-Family Loan Performance Data File Layout and Glossary. Each row contains 113 fields encoding static origination attributes, dynamic monthly performance measures, and post-termination expense and proceeds data. The estimation sample employed in this study draws on acquisition cohorts from 2013Q4 through 2022Q2, providing approximately nine years of origination vintages with performance histories spanning multiple interest-rate and housing-market cycles. Several earlier quarterly files were identified as incomplete during the data inventory stage and are excluded from the estimation sample.

## 1.2 Population and Eligibility

The primary dataset is not a census of all loans in Fannie Mae’s guarantee book; it is a curated research sample subject to documented eligibility and exclusion criteria fanniemae2024. The population is restricted to fixed-rate, fully amortising, full-documentation single-family conventional mortgage loans with original terms of at least five years and less than 35 years, originated on or after 1 January 1999 and acquired by Fannie Mae on or after 1 January 2000.

A number of product types are explicitly excluded. Adjustable-rate mortgages, interest-only loans, balloon-amortisation mortgages, and loans carrying prepayment penalties are absent from the dataset. Also excluded are loans with original loan-to-value (LTV) ratios exceeding 97 per cent, Alt-A and other reduced-documentation products, government-insured loans (FHA and VA), loans subject to long-term standby commitments or third-party risk-sharing arrangements other than primary mortgage insurance, and loans acquired on a negotiated bulk basis. Loans with non-standard payment due dates and loans that liquidated in the same month as acquisition are similarly excluded. Fannie Mae’s Home Affordable Refinance Program (HARP) loans are likewise absent from the primary dataset; when a loan in the primary dataset was subsequently refinanced through HARP, its termination is recorded as a standard prepayment, and its post-refinance performance is not tracked here. This is discussed further in Section ??.

These restrictions yield a sample of conventionally underwritten fixed-rate mortgages broadly representative of the agency conventional loan market under post-crisis underwriting standards. The exclusion of adjustable-rate and interest-only products limits the range of prepayment incentive dynamics captured relative to the full-spectrum dataset used by sirignano2021; this is discussed in Section ??. Table ?? summarises the principal eligibility criteria and exclusions.

## 1.3 Variables

The 113 fields in each monthly observation can be organised into three functional groups: static origination attributes, dynamic monthly performance attributes, and post-termination resolution data. The model uses a subset of these fields as covariates. Two fields — the Current Loan Delinquency Status and the Zero Balance Code — serve as the primary inputs from which the outcome variable is constructed.

| Criterion                     | Requirement / Exclusion                                                                     |
|-------------------------------|---------------------------------------------------------------------------------------------|
| Origination date              | On or after 1 January 1999                                                                  |
| Acquisition date              | On or after 1 January 2000                                                                  |
| Product type                  | Fixed-rate, fully amortising only                                                           |
| Loan term                     | $5 \leq \text{term} < 35$ years at origination                                              |
| Documentation                 | Full documentation only                                                                     |
| LTV at origination            | $\leq 97\%$                                                                                 |
| <i>Excluded: product</i>      | ARM, interest-only, balloon, prepayment-penalty loans                                       |
| <i>Excluded: credit class</i> | Alt-A, reduced-documentation, streamlined processing                                        |
| <i>Excluded: guarantee</i>    | Government-insured (FHA, VA) loans                                                          |
| <i>Excluded: structure</i>    | Long-term standby commitments; bulk acquisitions; lender-recourse loans (except primary MI) |
| <i>Excluded: programme</i>    | HARP and Refi Plus loans                                                                    |
| <i>Excluded: payment</i>      | Bi-weekly and non-first-of-month due dates                                                  |
| <i>Excluded: timing</i>       | Loans liquidated in month of acquisition                                                    |

Table 1.1: *Eligibility criteria for the Fannie Mae SFLP primary dataset.* Source: Fannie Mae Single-Family Loan Performance Dataset FAQs fanniemae2024.

### 1.3.1 Origination Attributes

Static origination attributes are recorded at loan delivery and do not change over the life of the loan absent a modification. The borrower’s FICO credit score at origination is available for both the primary borrower and, where applicable, a co-borrower; prior to the January 2015 dataset release only the lower of the two scores was disclosed, and a missingness indicator is constructed accordingly. Continuous origination covariates include the original debt-to-income (DTI) ratio, original LTV ratio, original combined LTV (CLTV, which incorporates subordinate liens), original unpaid principal balance (UPB), original interest rate, and original loan term in months. Categorical origination attributes include the acquisition channel (retail, broker, or correspondent), loan purpose (purchase, rate-term refinance, or cash-out refinance), property type, occupancy status (primary residence, second home, or investment property), property state, Metropolitan Statistical Area (MSA), and three-digit truncated zip code. The truncation of the zip code reflects borrower privacy considerations; full zip codes are not disclosed in the dataset fanniemae2024.

### 1.3.2 Dynamic Performance Attributes

Dynamic performance attributes are updated in each monthly reporting period. The current actual unpaid principal balance reflects the outstanding loan balance at each

point in time. Loan age, measured in months from the first payment date, and the remaining months to legal maturity are derived from the contractual amortisation schedule. The current interest rate reflects the applicable note rate, updated when a modification changes the loan terms.

Two dynamic fields jointly determine the outcome variable. The **Current Loan Delinquency Status** reports the number of missed monthly payments as a two-character code: 00 denotes a current loan; 01, 02, and codes 03 and above denote 30-, 60-, and 90-or-more-day delinquencies respectively; and XX denotes an unavailable status. The **Zero Balance Code** is populated exclusively in the terminal monthly observation and classifies the manner of loan termination into nine categories, distinguishing between voluntary prepayments, third-party foreclosure sales, short sales, repurchases, REO dispositions, note sales, and non-credit removals. The 24-month loan payment history string provides an additional source of behavioural information used to construct delinquency-recurrence and burnout features.

### 1.3.3 Externally Sourced Covariates

Two external series are appended to the loan-level panel to construct key time-varying covariates. The Freddie Mac Primary Mortgage Market Survey (PMMS) weekly fixed-rate series is used to compute the *rate incentive* for each loan-month observation, defined as the difference between the loan’s current interest rate and the prevailing market mortgage rate at the reporting period. This measure captures the refinancing option value and is the primary driver of prepayment behaviour in the empirical mortgage literature schwartz1989,sirignano2021. The Federal Housing Finance Agency (FHFA) House Price Index (HPI) at the MSA level is used to estimate the current LTV ratio by applying the index appreciation since origination to the original appraised value. Current LTV is the principal determinant of the default option value deng2000.

## 1.4 Outcome Variable

Following sirignano2021, the outcome variable is a categorical measure of mortgage state in month  $t + 1$ , conditional on the loan’s state and covariate vector in month  $t$ . Seven mutually exclusive states are defined: *Current*, *30 Days Past Due* (30 DPD), *60 Days Past Due* (60 DPD), *90+ Days Past Due* (90+ DPD), *Foreclosure*, *Real Estate Owned* (REO), and *Prepaid*.

State assignment for each loan-month observation follows a deterministic hierarchy. When the Zero Balance Code is present — indicating the terminal observation for the loan — the termination type is mapped to the Foreclosure, REO, or Prepaid absorbing state as shown in Table ???. When the Zero Balance Code is absent, the delinquency status field governs: codes 00 through 03 and above map to Current, 30 DPD, 60 DPD, and 90+ DPD respectively, while XX is treated as missing. The state in month  $t + 1$  is obtained by a window lead over the reporting period dimension, partitioned by loan identifier. Loans whose final observation falls at or before the data cut-off without a recorded termination event receive a censored indicator and are excluded from the labelled target.

Regarding HARP loans: as noted in Section ??, any loan in the primary dataset that was refinanced through the HARP programme terminates with a Zero Balance Code of 01 (prepaid) at the moment of refinancing and is therefore treated as a standard prepayment in the outcome variable. This approximation is defensible given that the loan is extinguished from Fannie Mae’s credit exposure in either case, though it may marginally overstate the rate-incentive sensitivity of the prepayment hazard during the period of peak HARP activity (approximately 2009–2013).

| State       | Source field       | Condition          | Economic interpretation                                                                       |
|-------------|--------------------|--------------------|-----------------------------------------------------------------------------------------------|
| Current     | Delinquency Status | = 00               | Borrower is contractually current on all payments.                                            |
| 30 DPD      | Delinquency Status | = 01               | One missed monthly payment.                                                                   |
| 60 DPD      | Delinquency Status | = 02               | Two consecutive missed monthly payments.                                                      |
| 90+ DPD     | Delinquency Status | $\geq$ 03          | Three or more missed payments; serious delinquency.                                           |
| Foreclosure | Zero Balance Code  | 02, 03, 15, 97, 98 | Credit-loss termination: third-party sale, short sale, note sale, or credit-event repurchase. |
| REO         | Zero Balance Code  | = 09               | Fannie Mae acquires ownership of the property.                                                |
| Prepaid     | Zero Balance Code  | 01, 06, 16, 96     | Full payoff, contractual maturity, repurchase, or non-credit removal (including HARP exits).  |

Table 1.2: *Mapping of Fannie Mae data fields to the seven-state outcome variable.* The Zero Balance Code is populated only in the terminal monthly observation; the Delinquency Status governs state assignment in all non-terminal periods.

The empirical monthly transition matrix derived from the estimation sample



serves as a natural in-sample benchmark against which the deep learning model is evaluated, mirroring the role of the empirical matrix in sirignano2021. As those authors document, transitions between non-adjacent delinquency states are non-trivial in practice: a meaningful fraction of seriously delinquent loans cure to current, and a small but non-zero share of loans in formal foreclosure eventually return to performing status. The seven-state formulation captures this richness in a way that a binary default-versus-prepayment classification would obscure.

## 1.5 Comparison to Sirignano et al. (2021)

The dataset employed in the present study differs in several important respects from the proprietary CoreLogic dataset used by sirignano2021, and these differences bear on the interpretation of the results.

sirignano2021 draw on records for approximately 120 million mortgages — both prime and subprime — originated across the United States between January 1995 and June 2014, yielding approximately 3.5 billion loan-month observations. The CoreLogic data encompass the full spectrum of mortgage products, including fixed-rate, adjustable-rate, hybrid, balloon, and interest-only loans, as well as government-insured instruments. Macro-economic covariates — zip-code-level house price indices from Zillow, county-level unemployment rates from the Bureau of Labor Statistics, and the Freddie Mac PMMS series — are merged at the loan-month level within the CoreLogic database. The dataset is proprietary and cannot be reproduced by independent researchers.

The Fannie Mae primary dataset, by contrast, is limited to conventional fixed-rate fully-amortising loans and excludes both subprime and government-insured products. Macro-economic covariates must be sourced and joined externally, as described in Section ???. These differences have two principal consequences. First, the estimation sample is more homogeneous in product mix; the nonlinear prepayment dynamics associated with adjustable-rate burnout and subprime option exercise — prominent in sirignano2021 — are absent. Second, the absence of a subprime segment limits the range of credit-risk outcomes observed; sirignano2021 report a median FICO score of 630 for their subprime sub-sample against 730 for the prime sub-sample, whereas the Fannie Mae population is concentrated at the higher end of this credit-quality spectrum.

These limitations notwithstanding, the Fannie Mae dataset provides a fully reproducible basis for the analysis. The core methodological contribution of sirignano2021

— applying deep learning to a multinomial competing-risks model of monthly loan-state transitions — transfers completely to the Fannie Mae panel. The architecture, estimation procedure, and evaluation framework described in subsequent chapters are direct adaptations of that approach. Table ?? summarises the key differences.

| Dimension               | Sirignano et al. (2021)         | Present study                           |
|-------------------------|---------------------------------|-----------------------------------------|
| Data source             | CoreLogic (proprietary)         | Fannie Mae SFLP (public)                |
| Total loans             | ~120 million                    | Tens of millions                        |
| Loan-month observations | ~3.5 billion                    | ~1.5–2 billion (estimation sample)      |
| Origination window      | 1995–2014                       | 1999–present; est. sample 2013–2022     |
| Products covered        | Fixed, ARM, hybrid, balloon, IO | Fixed-rate, fully amortising only       |
| Credit segments         | Prime and subprime              | Conventional (predominantly prime)      |
| Macro covariates        | Embedded (Zillow, BLS, PMMS)    | External join required (FHFA HPI, PMMS) |
| Geographic detail       | Full 5-digit zip code           | 3-digit truncated zip; MSA              |
| States modelled         | Seven                           | Seven (identical mapping)               |
| Reproducibility         | Not publicly reproducible       | Fully reproducible                      |

Table 1.3: *Comparison of the Fannie Mae SFLP primary dataset and the CoreLogic dataset used by sirignano2021.* Loan counts and observation counts are approximate. Estimation sample coverage for the present study reflects available acquisition quarters.